

PREAMBLE:

The faculty development programme is intended to bridge the gap between researchers/academia and the latest research facilities in Material characterization. This event exposes theoretical as well as actual experimental concept and knowledge required to conduct analysis and investigation pertaining to crystal structure, crystallite size, phase transformation, composition, grain boundary, glass transition behaviour, crystalline to amorphous transition, magnetic and mechanical properties at nano scale regime. In addition, this event showcases the importance of adopting these characterization techniques in industry.

OBJECTIVES:

- To help material researchers identify the kind of characterization techniques.
- To identify the structure and properties of unknown material.
- To investigate phase transformation, order / disorder transition and amorphous / crystalline structure in samples subjected to various deformation processes.
- To observe and measure grain size in nano scale regime.
- To impart the necessary technical knowledge and skills required to adopt different characterization techniques.
- To familiarize software used in different characterization processes.
- To bring about awareness of different characterization techniques to implement in industry.
- To create a continuous industry institute material characterization forum.
- To help young researchers identify state of the art centre of research excellence in material characterization.

TOPICS TO BE COVERED:

- X-Ray Diffraction Techniques.
- Electron microscopy techniques.
- Investigate hardness and elastic modulus in nanostructured samples using Nano indentation.
- Vibrating sample magnetometer to investigate saturation and remanence magnetization, coercivity and hysteresis loop.
- To investigate glass transition, recrystallisation and grain growth using differential scanning calorimeter.

RESOURCE PERSONS:

Eminent speakers from both academia and industry will address the participants in the workshop. Academia experts from IISc, NIT's and R&D organizations and experts in the field of Intellectual Property Rights and Patenting will deliver sessions.

WHO CAN ATTEND...?

Research scholars, academia and industry people especially in the areas of Mechanical Engineering, Material Science, Physics, Chemistry, Dental and Medical Sciences and Civil Engineering.

ABOUT THE DEPARTMENT:

Established in the year 1980, this department has dynamic, highly qualified and experienced faculty that is constantly engaged in R&D activities. Fully equipped labs with CAD/CAM/CAE and recent CNC training facilities and Condition Monitoring lab provides an environment where students can experience the vast expanse of Mechanical Engineering. The department has number of research projects in the areas of stress analysis, vibration, composite materials sponsored by AICTE / VTU. The department also offers M.Tech Degrees in Design Engineering and Computer Integrated Manufacturing and has been recognized as a research center by VTU, leading to Doctoral and M.Sc. (Engg.) degrees.

The department is conducting workshops for faculty & students, and special training in CAD/ CAM / CAE areas. The faculty has several papers published in national and international journal to their credit. Each year the department takes up assignments that focus on technical problems faced by industries. Students have won several awards for their projects and have been actively involved in the co-curricular & extra-curricular activities. The department is actively engaged in developing many innovative projects useful for the agricultural field and industries. The Automatic Sugarcane Seed Scooping machine developed got wide recognition and awards.

DEPARTMENT VISION:

To be a model department with academic excellence and technical expertise, grooming students to become Mechanical Engineers of best quality to meet ever changing needs of industry and society.

DEPARTMENT MISSION:

- Impart technical education of highest standard by qualified and experienced faculties in the field of Mechanical Engineering through effective teaching learning process.
- Continuously upgrade knowledge, skill and engage in Research & Development activities by executing, challenging projects in association with academia and industry.
- Imbibe ethical, environmental and social values amongst students using value based education.

National Education Society®

JAWAHARLAL NEHRU NATIONAL COLLEGE OF ENGINEERING, SHIVAMOGGA, KARNATAKA

(Affiliated to VTU Belagavi, Recognized by Govt. of Karnataka
Accredited by NAAC & Approved by AICTE, New Delhi)

Internal Quality Assurance Cell (IQAC)

www.jnnce.ac.in



Five Days National Level Workshop on MATERIAL CHARACTERIZATION

From 28th Jan - 1st Feb 2019

ORGANIZED BY

Department of Mechanical Engineering
JNNCE, Shivamogga

In association with ISTE, New Delhi

COORDINATOR

Dr. M. M. Rajath Hegde
Professor, Department of Mechanical Engineering

CONVENERS

Mr. Shanthi Kiran M.
Assistant Professor, Department of Mechanical Engineering

Mr. Jayavardhana Acharya N.
Assistant Professor, Department of Mechanical Engineering

Mr. Parameshwara S.
Assistant Professor, Department of Mechanical Engineering

Mr. Pradeep N. B.
Assistant Professor, Department of Mechanical Engineering

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- Sri T. R. Ashwathanarayana Setty, Vice President - NES
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- Dr. L. K. Sreepathi, Vice Principal, JNNCE, Shivamogga.

ADVISORY BOARD:

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IISc Bangalore & Founder Director, IIT, Punjab (Ropar)

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Dean and Professor, Department of Nano Science and Technology,
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Dr. Udaya Bhat K.

Professor, Materials & Metallurgical Department, NITK, Surathkal.

Dr. Vijay H. Desai

Professor, Mechanical Engineering Department, NITK, Surathkal.

ORGANIZING COMMITTEE:

- Dr. K. Sabeel Ahmed, HOD, Department of Mechanical Engineering, JNNCE.
- All Teaching and & Technical Staff of M.E. Dept.

COORDINATOR:

Dr. M. M. Rajath Hegde, Professor, Department of Mechanical Engineering, JNNCE.

ABOUT THE INSTITUTION:

National Education Society (NES) started in the year 1946 by a group of freedom fighters. Now it is running 40 educational institutions.

Jawaharlal Nehru National College of Engineering (JNNCE) was established by NES in 1980 and is one of the premier institutions in the central region of Karnataka and ranks within 10 in Karnataka. The college is catering the needs of the nearby as well as interstate students to pursue their graduation and post graduation in engineering, computer applications and business administration. Currently 7 UG and 9 PG programmes are being run successfully. The institution holds many milestones in research and innovation.

REGISTRATION FEE:

For faculties and research scholars : Rs. 1,000/-

For participants from industries : Rs. 2,000/-

Registration fee should be paid through online, the details of which are given below,

Bank : CANARA BANK

Branch : JNNCE CAMPUS, SHIMOGGA

Current A/c No. : 6322201000004

IFSC Code : CNRB0006322

Scanned copy of the application duly forwarded by the HOD & Principal, must be sent along with online payment details to Email address given below.

Participants will be intimated over email.

Last date for submission of application is 20th Jan 2019.

Email ID: characterization.material@gmail.com

CONTACT :

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Jayavardhana Acharya N. - +91 89712 84597

Parameshwara S. - +91 94806 48947

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NOTE:

Registration fee does not include accommodation. However, accommodation may be provided to the participants with a nominal fee in the college hostel. No TA/DA would be provided to participants.

ABOUT SHIVAMOGGA:

Shivamogga is a hill station situated in the mid South-Western Ghats of Karnataka state connected through National highway and railways from states capital Bangalore at a distance of 275 Km.

Shivamogga is a true nature's gift, spread with rivers, valleys, waterfalls, enchanting natural scenery of hills, dense forests, flora & fauna & forts. Among them Jog Falls is a world famous tourist attraction.

Five Days National Level Workshop on MATERIAL CHARACTERIZATION

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REGISTRATION FORM

Name: _____

Designation: _____

Department: _____

Address: _____

Email: _____

Mob. No. / Tel. No. / Fax No.: _____

Details of Online Payment:

Name of the Bank: _____

UTR No: _____

Accommodation Required:

YES ☐

NO ☐

Signature

Signature of the HOD